SHOPPER SPECTRUM: CUSTOMER SEGMENTATION AND PRODUCT RECOMMENDATIONS IN E-COMMERCE

ABSTRACT

The rapid growth of e-commerce has generated large volumes of customer purchasing data. Understanding this data is essential for providing personalized shopping experiences and improving business strategies. This project, **Shopper Spectrum**, uses machine learning techniques to segment customers based on their purchasing behavior and recommend suitable products.

The project performs data preprocessing, exploratory data analysis, feature engineering, hypothesis testing, and customer segmentation using the K-Means clustering algorithm. Based on the identified customer segments, personalized product recommendations are generated. An interactive web application is developed using Streamlit to allow users to predict customer segments and receive product recommendations. The application is deployed on Google Cloud Platform (Cloud Run), making it accessible through a web browser.

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CHAPTER 1: INTRODUCTION

E-commerce platforms collect huge amounts of customer data every day. Analyzing this information helps businesses understand customer preferences, improve marketing strategies, and enhance customer satisfaction. Customer segmentation is an important machine learning technique used to group customers with similar purchasing behavior. Recommendation systems further improve customer experience by suggesting products that match customer interests.

This project develops an intelligent recommendation system that combines customer segmentation using K-Means clustering with personalized product recommendations. A Streamlit web application provides an easy-to-use interface for users.

CHAPTER 2: LITERATURE SURVEY

Several researchers have proposed customer segmentation and recommendation systems to improve customer engagement in e-commerce. K-Means clustering is one of the most widely used unsupervised learning algorithms because of its simplicity and efficiency. Recommendation systems have become an essential component of modern online shopping platforms by helping customers discover relevant products.

CHAPTER 3: PROBLEM STATEMENT

Traditional e-commerce systems often provide generic recommendations that do not accurately reflect customer preferences. Without customer segmentation, businesses cannot effectively target marketing campaigns or improve customer satisfaction.

The objective of this project is to build a machine learning system that segments customers based on purchasing behavior and recommends suitable products.

CHAPTER 4: EXISTING SYSTEM

The existing system recommends products mainly based on popularity or recent purchases.

Limitations

- Generic recommendations
- No customer segmentation
- Lower personalization
- Less effective marketing

CHAPTER 5: PROPOSED SYSTEM

The proposed system performs:

- Data preprocessing
- Exploratory Data Analysis
- Customer segmentation using K-Means
- Product recommendation generation
- Interactive Streamlit interface
- Cloud deployment using Google Cloud Platform

CHAPTER 6: OBJECTIVES

- Analyze customer purchase data.
- Perform data preprocessing.
- Conduct exploratory data analysis.
- Apply K-Means clustering.
- Recommend products based on customer segments.
- Develop a Streamlit web application.
- Deploy the application on Google Cloud Platform.

CHAPTER 7: TECHNOLOGY STACK

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn

- SciPy
- Joblib
- Matplotlib
- Seaborn
- Streamlit

Tools

- Jupyter Notebook
- VS Code
- Git
- GitHub
- Docker
- Google Cloud Platform

CHAPTER 8: SYSTEM ARCHITECTURE

Customer Dataset

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Data Preprocessing

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Exploratory Data Analysis

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Feature Engineering

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K-Means Clustering

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Customer Segments

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Recommendation Engine

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Streamlit Web Application

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Google Cloud Platform

CHAPTER 9: METHODOLOGY

The project follows these steps:

1. Data Collection
2. Data Cleaning
3. Feature Engineering
4. Exploratory Data Analysis
5. Hypothesis Testing
6. K-Means Clustering
7. Product Recommendation
8. Streamlit Development
9. Cloud Deployment

CHAPTER 10: DATASET DESCRIPTION

The dataset contains customer information such as:

- Customer ID
- Gender
- Age
- Annual Income
- Spending Score
- Product Category
- Purchase Frequency
- Payment Method

CHAPTER 11: DATA PREPROCESSING

The following preprocessing techniques were applied:

- Missing value handling
- Duplicate removal
- Feature scaling
- Encoding categorical variables
- Outlier detection

CHAPTER 12: EXPLORATORY DATA ANALYSIS

EDA was performed using various visualizations including:

- Histograms
- Count plots
- Box plots
- Correlation heatmap
- Distribution plots

These visualizations helped understand customer behavior and relationships among variables.

CHAPTER 13: HYPOTHESIS TESTING

Hypothesis testing was performed to validate relationships between important customer attributes.

- Null Hypothesis (H_0): There is no significant relationship between selected variables.
- Alternative Hypothesis (H_1): There is a significant relationship between selected variables.

The test results supported the insights obtained during exploratory data analysis.

CHAPTER 14: CUSTOMER SEGMENTATION

The K-Means clustering algorithm was used to divide customers into groups based on purchasing behavior.

The Elbow Method was applied to determine the optimal number of clusters. Each cluster represents customers with similar characteristics, allowing businesses to implement targeted marketing strategies.

CHAPTER 15: PRODUCT RECOMMENDATION SYSTEM

After identifying customer segments, the system recommends products based on the purchasing patterns of customers within the same cluster.

This personalized recommendation approach improves customer satisfaction and increases sales opportunities.

CHAPTER 16: STREAMLIT WEB APPLICATION

A Streamlit web application was developed to provide an interactive user interface.

The application allows users to:

- Enter customer details
- Predict customer segments
- Display recommended products
- View prediction results instantly

CHAPTER 17: GOOGLE CLOUD PLATFORM DEPLOYMENT

The application was deployed using **Google Cloud Platform (Cloud Run)**.

Deployment steps included:

- Configuring the GCP project
- Enabling Cloud Run and Cloud Build services
- Creating a Dockerfile
- Defining project dependencies in `requirements.txt`
- Deploying the application using Cloud Run
- Making the application publicly accessible through its service URL

This deployment provides scalability, reliability, and easy accessibility over the internet.

CHAPTER 18: RESULTS

The project successfully:

- Segmented customers into meaningful groups.
- Generated personalized product recommendations.
- Developed an interactive web application.
- Successfully deployed the application on Google Cloud Platform.

CHAPTER 19: ADVANTAGES

- Personalized recommendations
- Improved customer understanding
- Better marketing decisions
- Easy-to-use interface
- Scalable cloud deployment

CHAPTER 20: LIMITATIONS

- Depends on dataset quality
- Sensitive to outliers
- Static recommendation logic
- Requires periodic model updates

CHAPTER 21: FUTURE SCOPE

- Deep Learning recommendation models
- Hybrid recommendation systems
- Real-time customer segmentation
- Mobile application
- AI-powered chatbot integration

CHAPTER 22: CONCLUSION

The Shopper Spectrum project demonstrates the effective use of machine learning in e-commerce. Customer segmentation using K-Means clustering enables businesses to understand customer purchasing behavior, while the recommendation system enhances user experience through personalized product suggestions. The deployment on Google Cloud Platform further makes the application accessible, scalable, and suitable for real-world use.
